How Sparse Are Useful Transformer Representations?

Random vs. Data-Driven Dimension Selection for Text Classification

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1 Introduction

1.1 Background and Motivation

Pretrained language models such as BERT [1] produce high-dimensional representations that support a wide range of downstream tasks. BERT-base, for example, generates dense, 768-dimensional embeddings. However, not all dimensions may be equally important for a particular prediction task. This motivates a question closely related to sparse inference: can a small subset of embedding coordinates preserve most of the predictive performance of the full representation?

Although BERT embeddings are dense, their predictive relationships with downstream targets may be sparse. Identifying informative subsets of coordinates can reveal task-specific predictive redundancy while reducing downstream storage and classifier computation. To investigate this, we compare supervised feature selection, random coordinate selection, and principal component analysis (PCA). Random selection helps assess whether predictive information is broadly distributed across coordinates, while supervised selection tests whether identifying task-relevant features offers an advantage. PCA provides a complementary benchmark by combining information across dimensions rather than selecting individual coordinates. Using frozen BERT embeddings, we examine how representation size and reduction strategy affect classification performance without modifying the underlying encoder.

1.2 Related Work

Previous studies have examined how linguistic information is distributed across neurons in pretrained language models. Durrani et al. [2] investigate morphology, syntax, and semantics and identify small neuron subsets capable of predicting linguistic properties. They find that morphological information is more localized than syntactic information, while information in BERT is more distributed than in XLNet. These findings suggest that the extent of predictive sparsity may depend on both the linguistic property and the underlying model.

Dalvi et al. [3] further investigate redundancy in pretrained Transformer representations, distinguishing general redundancy from task-specific redundancy. Their layer- and neuron-level analyses show that selected subsets can retain much of downstream predictive performance. Together, these studies suggest that high-dimensional representations may contain substantial predictive redundancy, although the extent to which dimensions can be removed depends on the task and representation.

Building on this work, we focus on a nar-

rower empirical question: how does the choice of dimension-reduction strategy affect predictive performance at a fixed representation size? Rather than examining linguistic properties across multiple layers, we study pooled BERT embeddings for sentiment classification. By comparing different reduction strategies at matched output dimensions, we examine whether preserving predictive performance depends on identifying task-relevant coordinates or can also be achieved through uninformed selection or projection.

Classical statistical methods provide complementary approaches to this comparison. The lasso introduced by Tibshirani [4] motivates using ℓ_1-regularized logistic regression for supervised coordinate selection. Principal component analysis (PCA) provides an unsupervised projection that preserves directions of maximum variance [5]. These methods differ in whether they use label information and whether they retain original coordinates or construct new components. Random coordinate selection provides an additional baseline without informed selection. Together, these approaches connect prior work on representation redundancy with statistical feature selection and dimension reduction.

1.3 Research Questions and Scope

We investigate three main questions. First, how much can the dimensionality of frozen BERT embeddings be reduced while maintaining classification performance close to the full-dimensional baseline? Second, how do supervised coordinate selection, random coordinate selection, and PCA compare at matched representation sizes? Third, how sensitive is classification performance to the particular coordinates retained under random selection? Together, these questions examine both the predictive sufficiency of reduced representations and the importance of how dimensions are selected.

We study these questions using frozen BERT-base embeddings for sentiment classification on Amazon Review Polarity [6]. Our contribution is an empirical comparison of established dimension-reduction methods, including an assessment of variability across random coordinate subsets. The findings are specific to this task and embedding basis. They do not establish universal sparsity of Transformer representations or imply reductions in the computational cost of BERT inference.

2 Data and Methodology

2.1 Dataset and preprocessing

We use Amazon Review Polarity, a binary sentiment dataset of product reviews, accessed through `fancyzhx/amazon_polarity` on Hugging Face.[1] One- and two-star reviews are labeled negative ($y = 0$), while four- and five-star reviews are labeled positive ($y = 1$). Three-star reviews are excluded.

We preserve the original training and test splits. To construct smaller samples, we first draw candidate pools of 60,000 training reviews and 30,000 test reviews without replacement using NumPy seed 0. We combine each review's title and content, adding a period after the title when needed, normalize whitespace, and remove empty texts and exact duplicates. We also remove test-pool texts that appear in the cleaned training pool. This removes exact text overlap but does not identify near-duplicates.

Using sampling seed 0, we draw balanced samples of 20,000 training reviews and 10,000 test reviews from the cleaned pools. Table 1 summarizes the samples. Word counts are based on whitespace-separated words.

Table 1: Summary of the preprocessed reviews.

Split	Reviews	Negative	Positive	Median words	95th pct. words
Train	20,000	10,000	10,000	70	161
Test	10,000	5,000	5,000	69	160

[1] `https://huggingface.co/datasets/fancyzhx/amazon_polarity`

2.2 Frozen BERT representation

We extract embeddings using `bert-base-uncased` [1], with all model parameters frozen. The model runs in evaluation mode with gradient computation disabled. Reviews are tokenized in batches of 64, padded within each batch, and truncated to a maximum of 256 tokens, including special tokens. Before truncation, median token counts are 91 for training reviews and 90 for test reviews. Only 0.17% of training reviews and 0.18% of test reviews exceed the limit.

For review i, we average the final-layer token vectors using the attention mask:

$$x_i = \frac{\sum_{t=1}^{T_i} a_{it} h_{it}}{\sum_{t=1}^{T_i} a_{it}} \in \mathbb{R}^{768},$$

where h_{it} is the token representation and a_{it} is its attention-mask value. Padding tokens are excluded from the average, while special tokens are included. The training and test embedding matrices have sizes $20{,}000 \times 768$ and $10{,}000 \times 768$, respectively. Extraction uses float16 autocasting on the GPU, and embeddings are stored as float32 arrays and reused across experiments.

2.3 Exploratory analysis

We examine training-review token lengths overall and by label. For the training embeddings, we summarize coordinate standard deviations, pairwise Pearson correlations, and correlations between individual coordinates and the sentiment label. We also examine cumulative explained variance from PCA on standardized training embeddings and label correlations of the first 50 component scores. These summaries describe how variation and marginal sentiment association are distributed across the representation. The EDA result will be presented in the result section.

2.4 Representation reduction

Before selection or projection, each embedding coordinate is standardized using its training-set mean and standard deviation. The same transformation is applied to the test embeddings. We compare 12 reduced representation sizes,

$$k \in \{1, 2, 5, 10, 25, 50, 75, \\ 100, 150, 200, 300, 400\},$$

with the full 768-dimensional baseline.

Full representation. The baseline retains all 768 coordinates and provides the reference for measuring performance losses.

Random coordinate selection. For each k, we select k coordinates uniformly without replacement and repeat this procedure 20 times. One NumPy random generator initialized with seed 42 produces the draws sequentially. Subsets are not constrained to be nested across sizes. A separate classifier is fitted for each subset.

Supervised L1 selection. We fit L1-regularized logistic regression to the standardized training embeddings using the `liblinear` solver, a maximum of 3,000 iterations, and random seed 42. The regularization path contains 33 logarithmically spaced values of the inverse regularization parameter C from 10^{-4} to 1.

For each k, we choose the smallest C, corresponding to the strongest penalty on the path, that yields at least k nonzero coefficients. We retain the k coordinates with the largest absolute coefficients. Every target size is reached in the full run; Table 2 gives the selector settings. The L1 model selects coordinates, and a separate L2 classifier produces the final predictions.

Table 2: L1 selector settings. Nonzero counts are measured before retaining the top k coordinates. Values of C are rounded.

k	Selector C	Nonzero coefficients
1	0.000316	6
2	0.000316	6
5	0.000316	6
10	0.000422	14
25	0.000750	27
50	0.00133	51
75	0.00316	88
100	0.00422	106
150	0.0100	183
200	0.0133	208
300	0.0316	333
400	0.0562	409

Principal component analysis. We fit PCA to the standardized training embeddings, retaining 400 components, and use the first k component scores for each reduced classifier. Training-derived centering and loadings are used to transform the test embeddings. We also record cumulative explained variance at each size.

PCA components combine information from the original coordinates. Equal k therefore means equal downstream input dimension, but does not imply equal dependence on the original coordinates or equal transformation cost.

2.5 Classifier and tuning

All representations use the same classifier pipeline: `StandardScaler` followed by L2-regularized logistic regression, with a maximum of 3,000 iterations. We tune the inverse regularization parameter C over 11 logarithmically spaced values from 10^{-4} to 10, using five-fold stratified cross-validation with shuffled folds and seed 42.

The value with the highest mean validation accuracy is selected, and the pipeline is refitted on the complete training set. Each configuration is evaluated on the same held-out test set, which is excluded from fitting and tuning. Representation selection and PCA are performed before classifier cross-validation, so the coordinate selection and PCA are not repeated within folds. Validation folds therefore influence the reduced representations, including through their labels for L1 selection. Cross-validation scores may consequently be optimistic. The test data remain excluded from these fitting steps.

2.6 Metrics and variability

We report test accuracy and macro-F1:

$$\text{Accuracy} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathbf{1}\{\hat{y}_i = y_i\},$$

$$\text{MacroF1} = \frac{F1_0 + F1_1}{2},$$

where $F1_c$ is the F1 score treating class c as the positive class. For either metric M, loss relative to the full representation is

$$\Delta M_m(k) = M_{\text{full}} - M_m(k),$$

where m denotes the reduction method. Positive values indicate lower performance. Differences in proportions are multiplied by 100 when reported in percentage points.

For random selection, we report the mean and sample standard deviation across the 20 subsets at each size. Figure bands show one standard deviation and describe variability across coordinate draws on fixed data, rather than uncertainty over new training or test samples. The full baseline, L1 selection, and PCA each have one fitted configuration per size; their variability across data samples is not estimated.

3 Results

3.1 Exploratory data analysis

Figure 1 shows that negative reviews are generally longer than positive reviews, with median lengths of 98 and 84 tokens, respectively. Only 0.17% of training reviews exceed the 256-token limit, so truncation affects a small fraction of the sample.

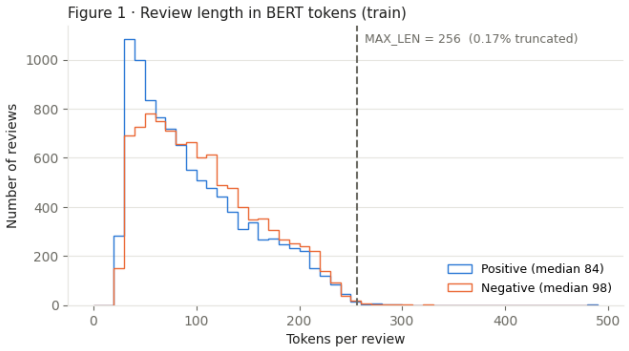


Figure 1: Training-review token lengths by sentiment. The dashed line marks the 256-token limit.

Figure 2 shows variation in coordinate scales, motivating standardization before selection and PCA. Most coordinate pairs have modest correlations, with a median absolute correlation of 0.084. Sentiment association varies across coordinates: 362 have absolute label correlations above 0.1, including 111 above 0.2. Thus, marginal sentiment association is spread across many coordinates, although its strength differs substantially.

Figure 3 shows that the first 100 principal components explain approximately 79% of standardized embedding variance, increasing to about 90% with 200 components. Among the first 50 components, PC4 has the strongest absolute label correlation, approximately 0.47. These results suggest that PCA can preserve substantial variation in fewer dimensions, while also showing that the highest-variance component is not necessarily the most associated with sentiment.

3.2 Classification performance

The full 768-dimensional representation achieves 90.80% accuracy and approximately 0.9080 macro-F1. Figure 4 shows that performance generally improves with representation size, with smaller gains at larger sizes. Accuracy and macro-F1 follow nearly identical patterns. Table 3 summarizes test accuracy.

Both L1 selection and PCA outperform average random selection at every tested size. Their relative performance depends on representation size. L1 selection performs better at $k = 1$ and $k = 2$, while PCA has higher observed accuracy at the tested sizes from $k = 5$ to $k = 75$. From $k = 100$ onward, L1 selection is slightly ahead.

At $k = 1$, L1 selection achieves 67.13% accuracy, compared with 56.84% for PCA and an average of 54.81% for random selection. At $k = 25$, L1 achieves 85.72% compared with 74.27% for random selection, an advantage of approximately 11.45 percentage points. These comparisons show that the reduction method matters particularly when few dimensions are retained.

3.3 Performance loss relative to the full representation

L1 selection preserves most of the baseline performance using a smaller coordinate subset. At $k = 100$, it retains approximately 13% of the coordinates and achieves 89.78% accuracy, a loss of 1.02 percentage points. At $k = 200$, it retains approximately 26% and achieves 90.46%, 0.34 percentage points below the baseline.

The gains diminish as more coordinates are retained. Increasing the L1 subset from 100 to 200 coordinates improves accuracy by 0.68 percentage points, whereas increasing it from 200 to 400 adds 0.19 points. L1 reaches 90.66% at $k = 300$ and 90.65% at $k = 400$, indicating little additional gain in this range. PCA also approaches the baseline, reaching 90.51% with 400 components. These close observed scores do not establish statistical equivalence.

3.4 Variability across random subsets

Average random-selection accuracy increases from 54.81% at $k = 1$ to 90.25% at $k = 400$. Variability is greatest at the smallest sizes: the standard deviation is 3.77 percentage points at $k = 1$ and 4.24 at $k = 2$. It generally decreases as more coordinates are retained, reaching 0.19 percentage points at $k = 400$.

Thus, small random subsets are sensitive to

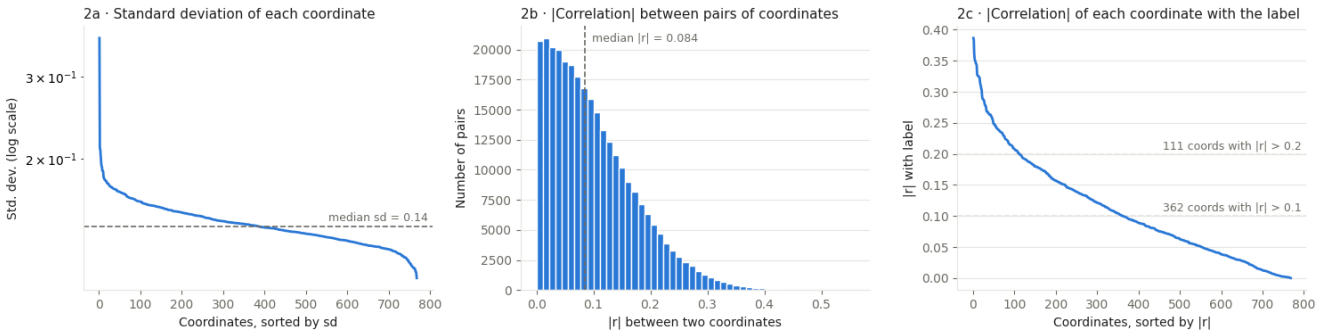


Figure 2: Training-embedding coordinate standard deviations, absolute pairwise correlations, and absolute correlations with sentiment labels.

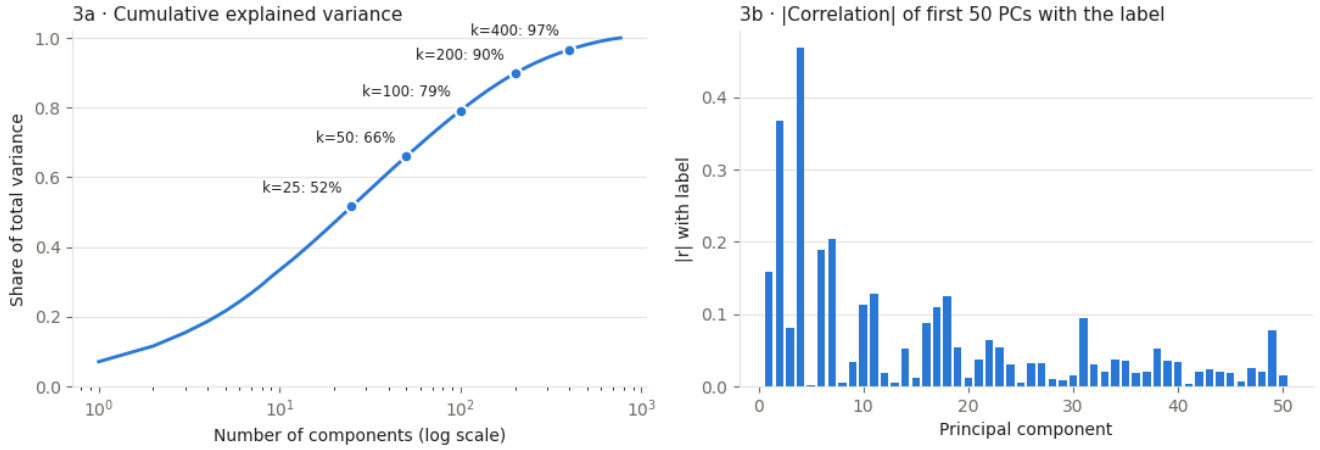


Figure 3: Cumulative explained variance from PCA on standardized training embeddings and absolute label correlations of the first 50 principal components.

which coordinates are selected. Larger subsets provide more consistent performance and approach the full baseline, although their average accuracy remains below L1 selection and PCA.

4 Discussion

4.1 Interpretation of the findings

The results show that a smaller representation can preserve much of the sentiment-classification performance of the full BERT embedding. However, the choice of retained coordinates matters. L1 selection outperforms average random selection at every tested size, particularly when few coordinates are retained. This suggests that supervised selection can identify coordinates that provide more useful predictive information than an arbitrary subset.

The EDA and classification results also suggest that distributed sentiment information and predictive redundancy can coexist. Many coordinates are individually associated with the label, yet a selected subset of 200 coordinates achieves 90.46% accuracy, compared with 90.80% for the full representation. This does not mean that the embedding vectors themselves are sparse or that the remaining coordinates contain no useful information. Their contributions may overlap with those of retained coordinates or be less useful to the linear classifier on this task. Coordinate selection also depends on the embedding basis, so the selected coordinates should not be interpreted as a universal set of sentiment features.

The expanded dimension grid reveals that neither L1 nor PCA consistently performs best.

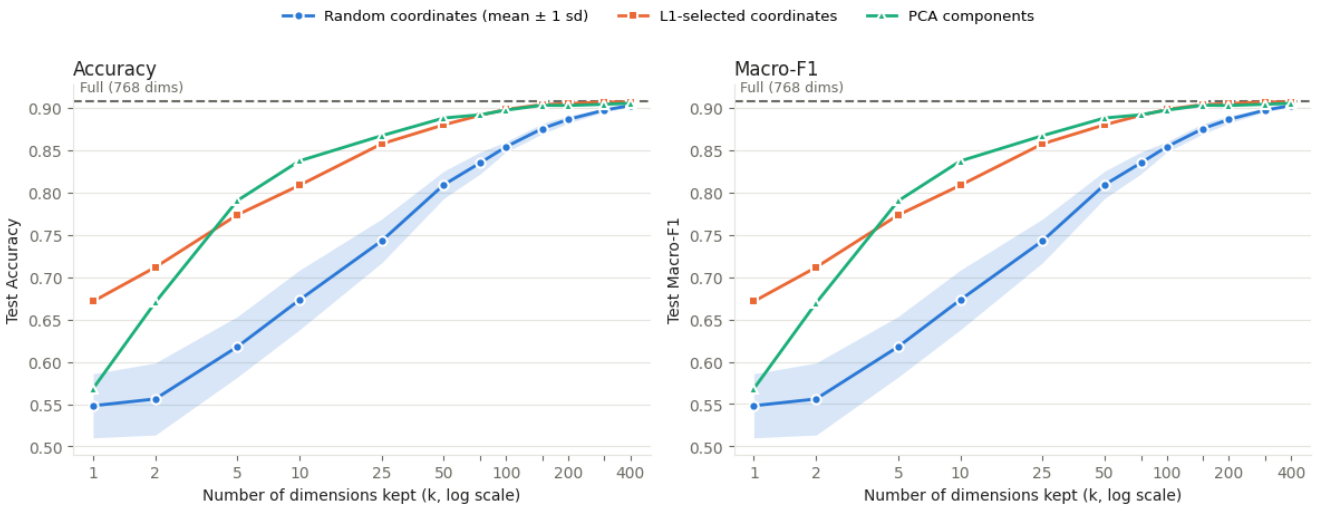


Figure 4: Test accuracy and macro-F1 versus representation size. The dashed line marks the full 768-dimensional baseline. Shaded bands show one standard deviation across 20 random coordinate subsets.

L1 is stronger at one and two dimensions, while PCA leads at the tested sizes from 5 to 75. L1 is slightly ahead at larger sizes. With one dimension, PCA achieves 56.84% accuracy compared with 67.13% for L1. This is consistent with the EDA observation that PC1 is not the component most associated with sentiment: preserving the greatest variance does not necessarily preserve the strongest predictive signal.

With more components, PCA can combine information from many coordinates and becomes competitive with supervised selection. At larger sizes, both methods approach the full baseline. The small remaining differences do not establish that either method is generally preferable.

4.2 Limitations and future work

The analysis uses one balanced sample from Amazon Polarity, one frozen BERT model, mean pooling, and a linear classifier. The findings concern this particular setup and may differ for other tasks, models, or pooling strategies. Exact-text deduplication also does not rule out near-duplicate reviews.

A methodological limitation is that L1 selection and PCA are fitted on the complete training set before classifier cross-validation. Validation folds therefore influence the representations used during tuning, making the validation scores potentially optimistic. The held-out test data remain excluded from fitting, but a cleaner procedure would refit standardization, selection, and PCA within each training fold.

The dimension grid was expanded after inspecting the initial results and evaluated on the same test set. Comparisons on the expanded grid should therefore be viewed as exploratory.

The reported random-selection variability measures sensitivity to coordinate choice on fixed data. It does not measure uncertainty across new data samples, and the L1 and PCA experiments do not assess selection stability across samples. Consequently, close test accuracies should not be interpreted as statistical equivalence or evidence that small differences are reproducible.

Finally, representation reduction occurs after the full BERT embedding is computed. It can reduce downstream feature storage and classifier size, but this implementation does not reduce transformer inference cost or establish end-to-end speed gains. Future work should first improve the cross-validation procedure and repeat the analysis across data samples. Additional tasks and paired comparisons of test predictions would help assess how broadly the findings hold and how uncertain the observed

Table 3: Test accuracy (%) by representation size. Random selection is reported as mean ± sample standard deviation in percentage points across 20 subsets. Bold values indicate the highest accuracy at each reduced size.

Dimensions (k)	Random selection	L1 selection	PCA
1	54.81 ± 3.77	**67.13**	56.84
2	55.61 ± 4.24	**71.15**	67.00
5	61.77 ± 3.57	77.34	**79.03**
10	67.30 ± 3.51	80.83	**83.71**
25	74.27 ± 2.57	85.72	**86.68**
50	80.89 ± 1.60	87.97	**88.77**
75	83.47 ± 1.28	89.08	**89.15**
100	85.37 ± 0.61	**89.78**	89.70
150	87.52 ± 0.60	**90.36**	90.28
200	88.60 ± 0.44	**90.46**	90.25
300	89.70 ± 0.20	**90.66**	90.39
400	90.25 ± 0.19	**90.65**	90.51
Full (768)	90.80		

performance gaps are.

5 Conclusion

Frozen BERT embeddings allow substantial representation reduction for Amazon Polarity sentiment classification. L1 selection retains 200 of 768 coordinates and achieves 90.46% accuracy, compared with 90.80% for the full representation. Both L1 selection and PCA outperform average random coordinate selection at every tested size. L1 performs better at one and two dimensions, PCA leads at the tested sizes from 5 to 75, and L1 is slightly ahead at larger sizes. These results support task-specific predictive redundancy and show that informed reduction preserves performance more effectively than random coordinate removal. They do not establish universal sparsity or an optimal dimension for transformer representations.

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A Implementation and Reproducibility

A.1 Software and execution settings

The experiments were implemented in Python using PyTorch, Transformers, Datasets, NumPy, pandas, and scikit-learn. The saved run records scikit-learn version 1.6.1 and CUDA-based embedding extraction. BERT inference uses float16 autocasting, while the resulting embeddings are stored as float32 arrays.

The reported experiment uses the complete sampled dataset, with `EMB_LIMIT=None` and `QUICK=False`. Data sampling uses seed 0, while the random coordinate generator, L1 solver, cross-validation splits, and PCA use seed 42. The random coordinate generator is initialized once and produces subsets sequentially across sizes and repetitions.

Data samples and embeddings are cached and reused when their files already exist. Changes to the dataset, model, or embedding settings therefore require regenerating the corresponding cached files.

A.2 Code and supporting materials

The implementation is provided in `stat236_embedding.ipynb`. Code and supporting materials are available at:

`https://github.com/leocwq/STAT236`

The notebook saves individual experiment results to `results/full/results_raw.csv` and aggregated performance to `results/full/summary.csv`. Embedding settings and token length statistics are stored in `embeddings/full/meta.json`. Exploratory summaries and figures are written to `results/full/eda/`.

An accompanying project webpage summarizes the research questions, methodology, and key findings:

`https://leocwq.github.io/STAT236/`

B Supplementary Results

B.1 Selected classifier hyperparameters

Table 4 reports the downstream L2 classifier's selected inverse regularization parameter C. These values are distinct from the L1 selector settings reported in the methodology section. Each representation is tuned separately using the same cross-validation procedure. The full 768-dimensional baseline selects $C = 0.01$.

Table 4: Selected downstream L2 classifier parameters. Values are rounded from the saved notebook outputs.

k	L1-selected features	PCA features
1	0.1	0.01
2	0.0001	0.000316
5	0.0001	0.000316
10	0.01	0.001
25	1	0.0316
50	0.01	10
75	1	0.00316
100	0.00316	0.00316
150	0.01	0.0316
200	0.00316	0.01
300	0.01	0.001
400	0.01	0.00316

B.2 Range of random-subset performance

Table 5 supplements the mean and standard deviation reported in the main results with the minimum and maximum accuracy across the 20 random subsets at each size. The observed range generally narrows as more coordinates are retained.

Table 5: Minimum and maximum test accuracy (%) across 20 random coordinate subsets at each size.

k	Minimum	Maximum
1	50.74	67.13
2	49.96	65.55
5	55.80	67.73
10	57.43	71.63
25	70.42	79.05
50	77.76	84.72
75	80.67	85.40
100	83.93	86.31
150	86.10	88.51
200	87.66	89.36
300	89.22	90.02
400	89.93	90.63

These ranges describe variation across coordinate subsets on the same training and test samples. They do not provide confidence intervals for performance on new samples.